

SHELL SCRIPTING

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manishaa@localhost:~$ # SHELL SCRIPTING COMMENTS - PART 1

#!/bin/bash
# This is a shebang. It tells the system to use Bash shell.

# 1. This script prints a greeting message
echo "Hello, Manishaa! Welcome to Shell Scripting." # Print greeting

# 2. Define a variable
name="Manishaa"      # String variable
age=22               # Integer variable

# 3. Use variables
echo "Name: $name"    # Print variable value
echo "Age: $age"      # Print variable value

# 4. Taking user input
read -p "Enter your city: " city # Read input from user
echo "You live in $city"         # Display user input

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manishaa@localhost:~$ # SHELL SCRIPTING COMMENTS - PART 2

# 1. Conditional Statement
if [ $age -ge 18 ]; then
    echo "You are an adult." # Condition is true
else
    echo "You are a minor."  # Condition is false
fi                             # End of if block

# 2. For Loop
echo "\nCounting from 1 to 5:" # Start of loop
for i in {1..5}; do
    echo "Number: $i"          # Loop body
done                          # End of loop

# 3. While Loop
count=1                       # Initialize counter
echo "\nWhile loop example:"
while [ $count -le 3 ]; do
    echo "Count: $count"      # Loop body
    ((count++))               # Increment counter
done                          # End of while loop

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manishaa@localhost:~$ # SHELL SCRIPTING COMMENTS - PART 3

# 1. Function Example
greet() {
    echo "Hello, $1! Have a great day!" # $1 is first argument
}                                         # End of function

greet "$name"                           # Function call

# 2. Check if a file exists
file="sample.txt"
if [ -f "$file" ]; then
    echo "$file exists." # File found
else
    echo "$file does not exist." # File not found
fi

# 3. Commenting multiple lines
: <<'END'                               # This is a multi-line comment
All lines between END markers
will be ignored by the shell.
Use this for big explanations.
END

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manishaa@localhost:~$ # SHELL SCRIPTING COMMENTS - PART 4

# 1. Case Statement
read -p "Enter your grade (A/B/C): " grade
case "$grade" in
    A) echo "Excellent!" ;; # Case A
    B) echo "Good job!" ;;  # Case B
    C) echo "Keep improving!" ;; # Case C
    *) echo "Invalid grade." ;; # Default case
esac                             # End of case

# 2. Command Substitution
current_date=$(date +%Y-%m-%d) # Store command output in variable
echo "Today's date is $current_date"

# 3. Exit Status
ls /not_exist_dir               # This will fail
if [ $? -ne 0 ]; then
    echo "Command failed! Check the directory."
fi                             # $? holds last command exit status

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